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Department of Electrical Engineering

Group Number	02

A PROJECT REPORT ON:

HYDROSENSE: Microcontroller based GSM water management system for real time monitoring and efficient usage control.

Submitted by

Project Team	
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In partial fulfilment of S.Y. B.Tech (Electrical Engineering)

EndSem of course

“Microcontroller and Integrated circuit based application ”

[ELPCC404 – Sem - IV]

Academic Year : 2023-24

CERTIFICATE

This is to certify that the project report entitled **“(HYDROSENSE) : Microcontroller based GSM water management system for real time monitoring and efficient usage control”**

Submitted by :

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is a bonafide work carried out by above students under the supervision of Mr. Sachin Shelar , and it is approved for the partial fulfilment of the end semester examination of course “Microcontroller and Integrated circuit based application”[ELPCC404- Sem - IV] of Second Year B. Tech Electrical Engineering program.

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Head of Department
Department of Electrical Engineering

External Examiner

Date: 05/06/24

Place: Pune

Abstract

This project is a microcontroller-based GSM water management system designed to facilitate real-time monitoring, efficient usage control, and support sustainable development of water resources. Water scarcity is a growing concern globally, necessitating the development of innovative solutions for sustainable water management. The proposed system integrates a microcontroller with GSM technology to provide a comprehensive solution for managing water supply and consumption.

The system employs sensors to monitor various parameters, such as water level, flow rate, and quality, in real-time. These sensors feed data to the microcontroller, which processes and transmits the information via GSM modules to a central server or directly to user's mobile devices. This real-time data transmission enables users and authorities to make informed decisions regarding water distribution and usage.

Additionally, the system includes automated control mechanisms to regulate water supply based on preset thresholds and user-defined parameters, ensuring optimal water utilization and preventing wastage. Users can also receive alerts and updates through SMS notifications, providing them with actionable insights to manage their water consumption effectively.

A critical component of the system is its contribution to sustainable development. By enabling precise control and monitoring of water resources, the system supports the conservation of water, promotes equitable distribution, and helps in mitigating the impacts of water scarcity. This approach aligns with global sustainable development goals by fostering responsible water management practices and enhancing resilience to water-related challenges.

The implementation of this system demonstrates significant potential in reducing water wastage, enhancing resource allocation, and promoting sustainable water usage practices. The paper discusses the design, development, and testing of the system, highlighting its effectiveness and scalability for deployment in various applications, from household water management to large-scale agricultural and industrial water distribution networks.

This report details the design, development, and implementation of the microcontroller-based GSM water management system. It highlights the system's effectiveness in various applications, from residential water management to large-scale agricultural and industrial water distribution networks. By demonstrating significant potential in reducing water wastage and promoting sustainable usage practices, the proposed system represents a critical advancement in the field of water resource management.

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Table of Contents

Chapter : 1 Introduction

Water is an indispensable resource, essential for sustaining life, agriculture, and industry. However, the increasing global demand for water, driven by rapid urbanization, population growth, and the impacts of climate change, has heightened the urgency for more efficient and sustainable water management practices. Traditional water management systems are often hampered by a lack of real-time monitoring capabilities and sophisticated control mechanisms, leading to significant inefficiencies and wastage.

In response to these challenges, this paper proposes a microcontroller-based water management system enhanced with Global System for Mobile Communications (GSM) technology. This innovative system is designed to facilitate real-time monitoring, precise control, and sustainable management of water resources. By leveraging the capabilities of microcontrollers and GSM technology, the system offers a robust solution for optimizing water usage and mitigating wastage.

The proposed system integrates ariouss sensors to continuously monitor key parameters such as water levels, flow rates, and quality metrics. Data from these sensors are processed by a central microcontroller, which then communicates the information via GSM modules to a central server or directly to users' mobile devices. This real-time data transmission enables users and authorities to make data-driven decisions regarding water distribution and consumption, thereby enhancing the overall efficiency of water management processes.

A key feature of this system is its contribution to sustainable development. By enabling precise monitoring and control, the system supports the conservation of water resources, ensures equitable distribution, and addresses the pressing issues of water scarcity. This aligns with the United Nations Sustainable Development Goals (SDGs), particularly Goal 6, which aims to ensure availability and sustainable management of water and sanitation for all.

This report details the design, development, and implementation of the microcontroller-based GSM water management system. It highlights the system's effectiveness in various applications, ranging from residential water management to large-scale agricultural and industrial water distribution networks. By demonstrating significant potential in reducing water wastage and promoting sustainable usage practices, the proposed system represents a critical advancement in the field of water resource management.

Chapter : 2 Problem statement & Objectives

2.1. Problem statement

- **No Real-Time Monitoring:** Existing systems can't provide immediate data on water parameters, delaying decisions.
- **Inefficient Control:** Without automation, water distribution is often mismanaged.
- **High Wastage:** Poor monitoring and control result in significant water wastage.
- **Low User Awareness:** Users lack insights into their consumption, hindering conservation efforts.
- **Sustainability Issues:** Current methods don't support sustainable water management or align with global development goals.

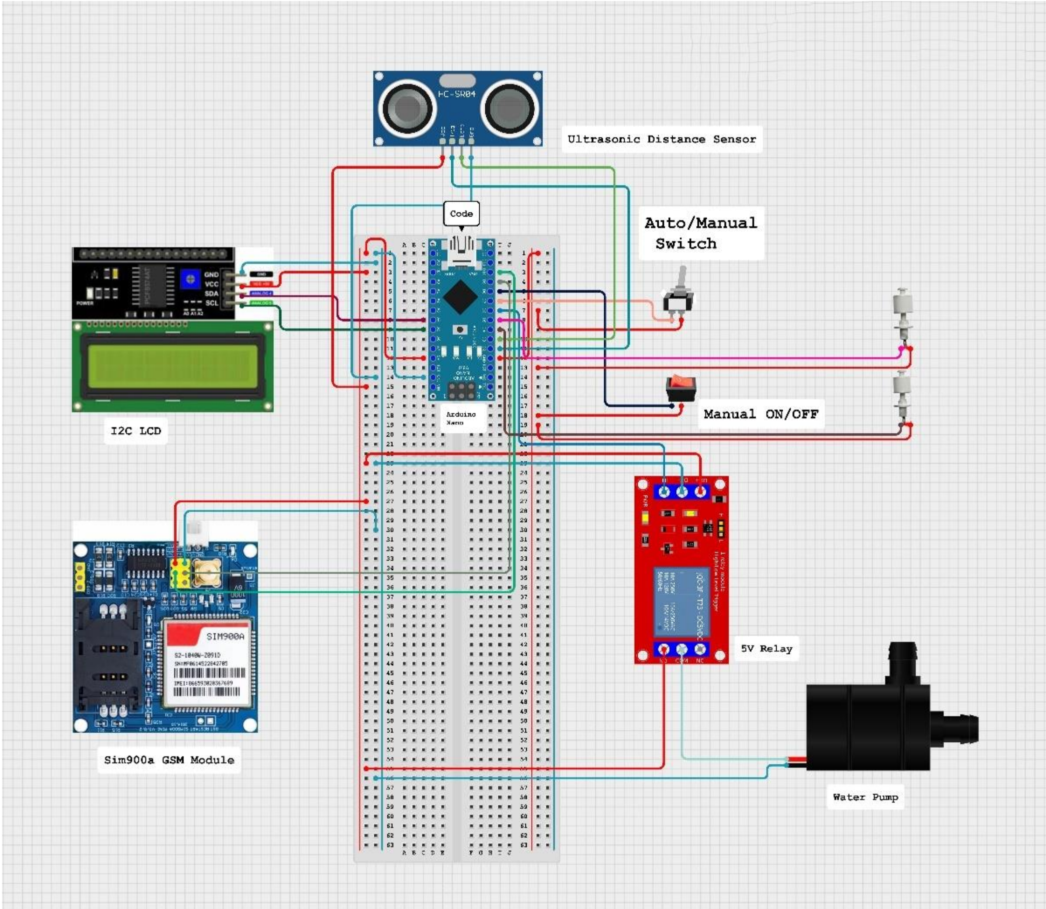
2.1. Objectives

- To Implement sensors to continuously monitor water levels, flow rates, and quality parameters.
- To Automated Control by developing automated mechanisms to regulate water distribution based on real-time data.
- To Utilize GSM technology to transmit real-time data to a central server and users' mobile devices.
- To Provide timely alerts and updates to users via SMS.
- Reduce water wastage by optimizing the control and distribution of water resources.
- To support sustainable water management practices that align with global development goals.
- To Increase the resilience of communities and industries to water scarcity and related challenges through efficient water management.

Chapter : 3 Hardware/Software Design

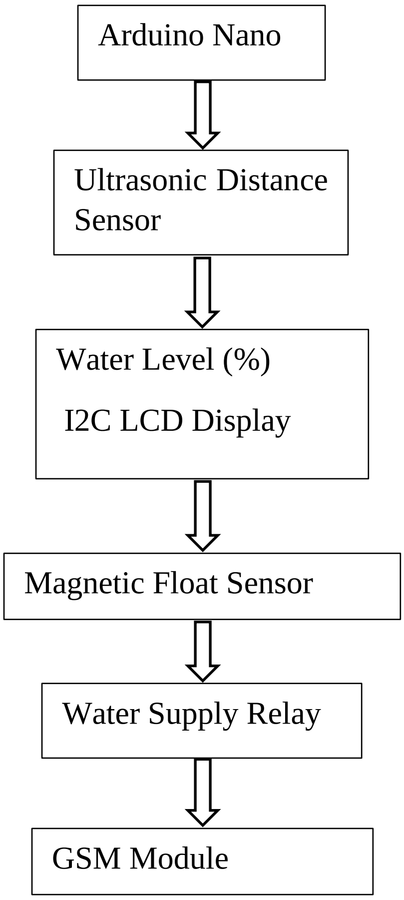
3.1. Introduction to project

3.1.1 Circuit Diagram



Circuit Diagram Microcontroller based GSM water management system for real time monitoring and efficient usage control.

3.1.2 Block diagram



3.1.3 Explanation Of Block Diagram

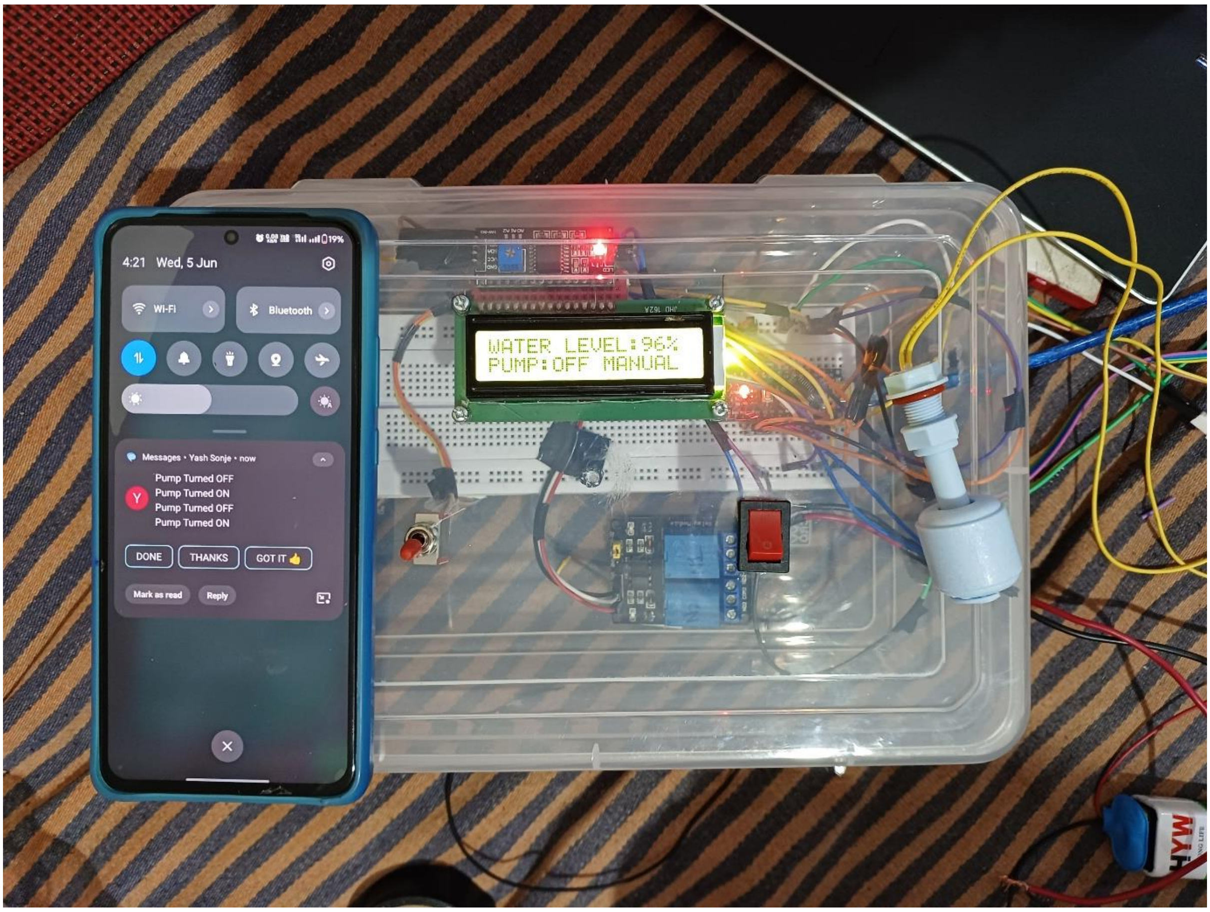
- **Arduino Nano**: The central microcontroller that manages all operations and communication within the system.
- **Ultrasonic Distance Sensor**: Measures the water level inside the tank and sends distance data to the Arduino Nano. The Arduino processes this data to calculate the water level percentage.
- **I2C LCD Display**: Displays the current water level percentage for real-time monitoring and user interface interaction.
- **Magnetic Float Sensor**: Monitors the water level in the tank. When the water level reaches specific thresholds, it sends signals to the Arduino Nano to control the water supply.
- **Water Supply Relay**: Actuates the water supply based on signals from the magnetic float sensor and Arduino Nano. It controls whether to allow water to flow into the tank or stop the supply.
- **GSM Module**: Enables real-time monitoring and alerts. It communicates with the Arduino Nano to send notifications via SMS when the tank is full, empty, or when certain conditions are met.

This configuration ensures efficient water management by continuously monitoring the water level, controlling supply based on predefined thresholds, and providing real-time alerts for effective usage control. Adjustments can be made based on specific operational requirements and environmental conditions.

3.1.4 Outcome


4:20 Wed, 5 Jun
Wi-Fi
Bluetooth
Messages • Yash Sonje • now
Y
Pump Turned OFF
Pump Turned ON
Pump Turned OFF
Pump Turned ON
DONE
THANKS
GOT IT
Mark as read
Reply
WATER LEVEL:96%
PUMP:ON AUTO


4:21 Wed, 5 Jun
Wi-Fi
Bluetooth
Messages • Yash Sonje • now
Pump Turned OFF
Pump Turned ON
Pump Turned OFF
Pump Turned ON
Pump Turned OFF
DONE
GOT IT
THANKS
Mark as read
Reply
WATER LEVEL:96%
PUMP:ON MANUAL


4:21 Wed, 5 Jun
Wi-Fi
Bluetooth
Messages • Yash Sonje • now
Pump Turned OFF
Pump Turned ON
Pump Turned OFF
Pump Turned ON
DONE
THANKS
GOT IT
Mark as read
Reply
WATER LEVEL:96%
PUMP:OFF MANUAL

3.2. Selection of components

3.2.1 Component 1 : Ultrasonic distance sensor .

An ultrasonic distance sensor measures the distance to an object using ultrasonic sound waves. It consists of a transmitter that emits sound waves and a receiver that detects the waves after they bounce back from the object. The time taken for the waves to return is used to calculate the distance. The working principle of an ultrasonic distance sensor involves four key steps. First, the transmitter emits a burst of ultrasonic sound waves. These sound waves travel through the air and reflect off an object. Upon reflection, the receiver detects the returning sound waves. The control circuit then calculates the distance to the object by measuring the time difference between the emission and reception of the sound waves.in this project , ultrasonic sensor is used to detect the level of water .

3.2.2 Component 2 : magnetic floating sensor .

A magnetic float sensor is a type of level sensor commonly used to detect the level of liquid within a container or tank. It operates on the principle of buoyancy and magnetism, where a float containing a magnet moves up and down along a stem as the liquid level changes. Magnetic float sensors come in two main types: open switch and closed switch.

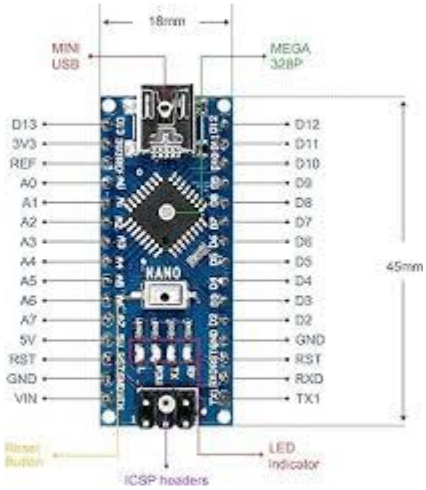
Open Switch Type: In an open switch magnetic float sensor, the reed switch remains open when the float is at its lowest position. As the liquid level rises, the float also rises, eventually reaching a point where the magnet within the float activates the reed switch, closing the circuit. This closing of the circuit indicates that the liquid level has reached a predetermined point.

Closed Switch Type: In a closed switch magnetic float sensor, the reed switch remains closed when the float is at its lowest position. As the liquid level rises, the float rises with it. When the float reaches a certain point, the magnet within the float moves away from the reed switch, opening the circuit. This opening of the circuit indicates that the liquid level has reached the predetermined point.

Both types of magnetic float sensors are commonly used in various applications for liquid level monitoring and control. The choice between open switch and closed switch types depends on the specific requirements of the application, such as the desired behavior of the sensor when the liquid level reaches a certain point.

3.2.3 Component 3: Arduino Nano

The Arduino Nano is a compact and versatile microcontroller board based on the ATmega328P chip. It shares many similarities with the Arduino Uno but comes in a smaller form factor, making it ideal for projects with space constraints. The Nano features a wide range of digital and analog input/output pins, allowing for versatile connectivity with sensors, actuators, and other electronic components. It can be programmed using the Arduino IDE, which offers a user-friendly interface for writing and uploading code. With its USB interface for programming and serial communication, the Nano simplifies the development process for hobbyists, students, and professionals alike. Despite its small size, the Arduino Nano packs a punch, offering the same capabilities as larger Arduino boards in a compact package.



3.2.4 Component 4: Relay

A single-channel relay is a type of electromechanical switch that controls the operation of a single electrical circuit. It consists of a coil of wire that, when energized, generates a magnetic field, causing a switch mechanism to open or close. Single-channel relays are commonly used in applications where the control of a single high-power electrical device, such as a motor, light, or heater, is required. They offer isolation between the control signal and the load, protecting the controlling circuit from potential damage due to high voltage or current. Single-channel relays typically have two states: normally open (NO) and normally closed (NC), where the state changes depending on whether the relay coil is energized or de-energized. These relays are widely used in various industrial, automotive, and home automation applications, providing a reliable and efficient means of controlling electrical devices remotely or automatically.

3.2.5 Component 5: LCD with I2C module

An LCD (Liquid Crystal Display) with an I2C (Inter-Integrated Circuit) module combines the functionality of a standard LCD display with the convenience of I2C communication protocol. The LCD provides a visual interface for displaying text and graphics, while the I2C module simplifies the connection and communication between the LCD and microcontroller, such as the Arduino Nano. With I2C, multiple devices can communicate over a two-wire serial interface, reducing the number of pins needed for connection. This integration streamlines the wiring process and conserves valuable GPIO pins on the microcontroller. Additionally, the I2C module typically includes an onboard potentiometer for adjusting the contrast of the LCD display, enhancing readability. LCDs with I2C modules are commonly used in embedded systems, IoT devices, and DIY projects, where space constraints or simplicity in wiring are considerations. They offer a user-friendly and efficient solution for displaying information in various applications, ranging from temperature and humidity monitoring to real-time data logging and menu-driven interfaces.

3.2.6 Component 6: Water Pump

A mini water pump is a compact and efficient device designed to circulate or transfer water in small-scale applications. Typically powered by electricity, these pumps are characterized by their small size, low power consumption, and often, submersible capabilities. Mini water pumps are commonly used in various contexts, including aquariums, hydroponic systems, small fountain displays, and DIY projects. They come in different configurations, such as centrifugal, diaphragm, or submersible pumps, each suited to specific applications based on factors like flow rate, pressure requirements, and fluid compatibility. Despite their compact size, mini water pumps can provide adequate water flow for their intended purposes, making them a versatile and practical solution for tasks requiring water circulation or transfer in confined spaces.

3.3. Hardware / Software

Software :

- **Arduino IDE**

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them.programs written using Arduino Software (IDE) are called sketches**.** These sketches are written in the text editor and are saved with the file extension .ino. The editor has features for cutting/pasting and for searching/replacing text. The message area gives feedback while saving and exporting and also displays errors. The console displays text output by the Arduino Software (IDE), including complete error messages and other information. The bottom righthand corner of the window displays the configured board and serial port. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

- **Proteus**

Proteus is a software tool used for electronic design automation (EDA). It is primarily used for designing, simulating, and testing electronic circuits before they are physically implemented. Proteus is popular among electronics engineers, students, and hobbyists for its ease of use and comprehensive set of features.

Key features of Proteus software include:

Schematic Capture: Proteus allows users to draw circuit diagrams using a graphical interface. Components such as resistors, capacitors, transistors, and integrated circuits can be placed and connected to simulate the circuit's functionality.

Simulation: Proteus includes a powerful simulation engine that can simulate the behavior of both analog and digital circuits. Users can test their designs for functionality, performance, and correctness before moving on to the physical implementation.

PCB Design: Proteus can also be used for designing printed circuit boards (PCBs). Users can transfer their circuit designs from the schematic capture tool to the PCB layout tool for designing the physical layout of the circuit on a PCB.
Microcontroller Simulation: Proteus supports simulation of microcontroller-based circuits. It includes a wide range of microcontroller models from various manufacturers, allowing users to develop and test embedded systems designs.
Virtual Instrumentation: Proteus includes a suite of virtual instruments that can be used to interact with the simulated circuits. Users can add oscilloscopes, function generators, logic analyzers, and other instruments to observe and analyze the circuit's behavior.
Code Debugging: For microcontroller-based designs, Proteus allows users to debug the firmware code by simulating the execution of the code step by step and observing the effects on the circuit.
Overall, Proteus is a versatile tool for electronic circuit design and simulation, offering a comprehensive set of features to support the entire design process from concept to implementation.

- **Cirkit Designer**

Cirkit Designer is a powerful and intuitive circuit design software that enables users to create and simulate digital electronic circuits with ease. The software provides a user-friendly interface that allows users to drag and drop components onto a virtual canvas, and then connect them together using virtual wires.

With Cirkit Designer, users can create complex circuits with a wide range of components, including logic gates, counters, registers, and more. The software also includes a range of virtual instruments for testing and debugging circuits, such as oscilloscopes, logic analyzers, and signal generators.

One of the key features of Cirkit Designer is its ability to simulate circuit behavior, allowing users to test and debug their designs without the need for physical prototyping. This saves time and reduces the risk of errors, making it an ideal tool for both hobbyists and professionals.

Cirkit Designer also includes a range of advanced features, such as signal analysis and waveform visualization, making it an ideal tool for anyone working with digital electronic circuits.

- **Code**

```
#include <EEPROM.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SoftwareSerial.h>

// LCD
LiquidCrystal_I2C lcd(0x27, 16, 2);

// GSM
SoftwareSerial gsmSerial(9, 10); // RX, TX for GSM SIM900A

// Mobile number to send SMS
const char phone_number[] = "+918308278406";

// To track relay state changes
bool relayState = false;
bool lastRelayState = false;

// Ultrasonic sensor and water level
long duration, inches;
int set_val, percentage;
bool state;

// Function prototypes
void measureWaterLevel();
void updatePercentage();
void displayInfo();
void sendSMS(const char* message);
```

```
long microsecondsToInches(long microseconds);

void setup() {
  // Initialize LCD
  lcd.init();
  lcd.backlight();
  lcd.print("WATER LEVEL:");
  lcd.setCursor(0, 1);
  lcd.print("PUMP:OFF MANUAL");

  // Configure pin modes for the ultrasonic sensor
  pinMode(2, OUTPUT); // Trigger pin of ultrasonic sensor
  pinMode(3, INPUT);  // Echo pin of ultrasonic sensor

  // Read set value from EEPROM
  set_val = EEPROM.read(0);
  if (set_val > 150) set_val = 150;

  // Configure pin modes for the GSM and relay control
  pinMode(4, INPUT_PULLUP); // Input pin 1
  pinMode(5, INPUT_PULLUP); // Input pin 2
  pinMode(6, OUTPUT);      // Relay output
  pinMode(7, INPUT_PULLUP); // Auto/Manual mode switch
  pinMode(8, INPUT_PULLUP); // Manual relay control

  // Initialize the relay as off
  digitalWrite(6, LOW);

  // Initialize GSM module
```

```
  gsmSerial.begin(9600);
  delay(1000);
  gsmSerial.println("AT");
  delay(1000);
  gsmSerial.println("AT+CMGF=1"); // Set SMS mode to text
  delay(1000);
}

void loop() {
  // Read the states of pin 4, pin 5, pin 7, and pin 8
  int input1 = digitalRead(4);
  int input2 = digitalRead(5);
  int modeSwitch = digitalRead(7);
  int manualControl = digitalRead(8);

  // Determine the relay state based on the mode and input states
  if (modeSwitch == HIGH) { // Auto mode
    if (input1 == HIGH && input2 == HIGH) {
      // Both inputs are HIGH
      digitalWrite(6, HIGH); // Turn on the relay
      relayState = true;
    } else {
      // Either input1 is HIGH or both are HIGH
      digitalWrite(6, LOW); // Turn off the relay
      relayState = false;
    }
  } else { // Manual mode
    if (manualControl == HIGH) {
      // Manual control is HIGH
```

```
    digitalWrite(6, HIGH); // Turn on the relay
    relayState = true;
  } else {
    // Manual control is LOW
    digitalWrite(6, LOW); // Turn off the relay
    relayState = false;
  }
}

// Check for relay state change to send SMS
if (relayState != lastRelayState) {
  if (relayState) {
    sendSMS("Pump Turned OFF");
  } else {
    sendSMS("Pump Turned ON");
  }
  lastRelayState = relayState;
}

// Measure water level
measureWaterLevel();

// Update percentage
updatePercentage();

// Display information on LCD
displayInfo();

delay(1000); // Wait for 1 second
```

```
}

void measureWaterLevel() {
  digitalWrite(2, HIGH);
  delayMicroseconds(10);
  digitalWrite(2, LOW);
  duration = pulseIn(3, HIGH);
  inches = microsecondsToInches(duration);
}

void updatePercentage() {
  percentage = (set_val - inches) * 100 / set_val;
  if (percentage < 0) percentage = 0;
}

void displayInfo() {
  lcd.setCursor(12, 0);
  lcd.print(percentage);
  lcd.print("%   ");

  lcd.setCursor(5, 1);
  if (relayState) lcd.print("OFF ");
  else lcd.print("ON ");

  lcd.setCursor(9, 1);
  if (digitalRead(7)) lcd.print("AUTO   ");
  else lcd.print("MANUAL");
}
```

```
long microsecondsToInches(long microseconds) {
  return microseconds / 74 / 2;
}

void sendSMS(const char* message) {
  gsmSerial.print("AT+CMGS=\"");
  gsmSerial.print(phone_number);
  gsmSerial.println("\"");
  delay(1000);
  gsmSerial.print(message);
  gsmSerial.write(26); // CTRL+Z to send
  delay(1000);
}
```

- **Code explanation**

This code appears to be written for an Arduino microcontroller and controls a water pump based on water level and user input. Let's break down the code section by section:

Libraries and Pin Setup:

- The code includes several libraries for functionalities like LCD display, GSM communication, and ultrasonic sensor interaction.
- It defines pins for the LCD, GSM module (SIM900A), ultrasonic sensor, relay control, and various input/output pins.
- It initializes the LCD and GSM module, setting communication parameters.

Variables:

- Relay State and last Relay State track the current and previous state of the relay (pump).
- duration stores the time it takes for the ultrasonic pulse to travel to the water surface and back.
- inches calculates the water level distance based on the duration.
- Setval stores the desired water level threshold read from EEPROM memory.
- percentage represents the water level remaining as a percentage.
- state (not used extensively in the code provided) might be a placeholder for future functionalities.

Functions:

- measureWaterLevel: Triggers an ultrasonic pulse, measures the echo time, and converts it to water level distance in inches.
- updatePercentage: Calculates the percentage of water remaining based on the set threshold and measured distance.
- displayInfo: Updates the LCD screen with the water level percentage, pump state (ON/OFF), and control mode (AUTO/MANUAL).
- sendSMS: Sends an SMS notification to a specified phone number when the pump state changes (turns ON or OFF).
- microsecondsToInches: Converts the measured echo time in microseconds to water level distance in inches.

Main Loop (loop() function):

- Reads the states of multiple input pins used for control logic (potentially buttons or switches).

- Determines the relay state based on the selected mode (AUTO or MANUAL) and input conditions.
- Sends SMS notification if the relay state changes.
- Calls functions to measure water level, update percentage, and display information on the LCD.
- Includes a delay of 1 second between loop iterations.

Overall Functionality:

This code automates a water pump based on the water level. In AUTO mode, the pump turns on when two specific input conditions are met (likely indicating low water level) and turns off when those conditions are not present. In MANUAL mode, a separate input pin directly controls the pump state (ON/OFF). The system also displays the water level percentage on the LCD and sends SMS alerts when the pump turns on or off.

Chapter : 4 Costing

Following table gives the costing of the project.

Sr. No.	Specification/Description of component	Quantity	Unit Cost (`)	Total (`)
1.	Arduino Nano	1	**₹ 380**	**₹ 380**
2.	Ultrasonic Distance Sensor	1	**₹ 100**	**₹ 100**
3.	Relay		**₹ 110**	**₹ 110**
4.	LCD & I2C module	1	**₹ 240**	**₹ 240**
5.	Magnetic float sensor	1	**₹ 175**	**₹ 175**
6.	Sim 900a gsm module	1	**₹ 850**	**₹ 850**
7.	Rocker switch	1	**₹ 20**	**₹ 20**
8.	Toggle switch	1	**₹ 50**	**₹ 50**
9.	Jumper wires	20	**₹ 3**	**₹ 60**
10.	Bread Board	1	**₹ 50**	**₹ 50**
			total	**₹ 2,035**

Chapter : 5 Member work sheet

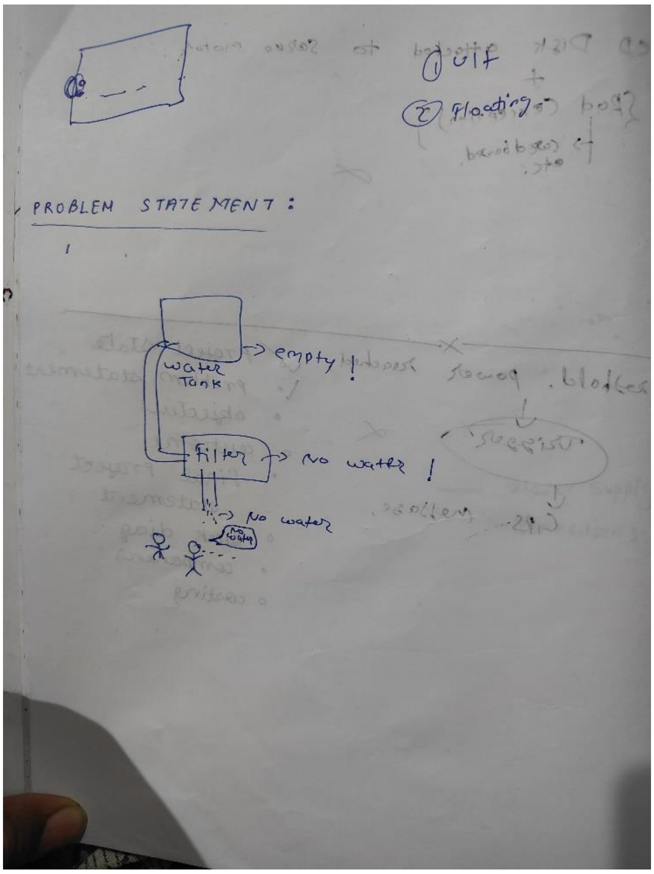
Following table gives the details of the work allotted and completed by each member.

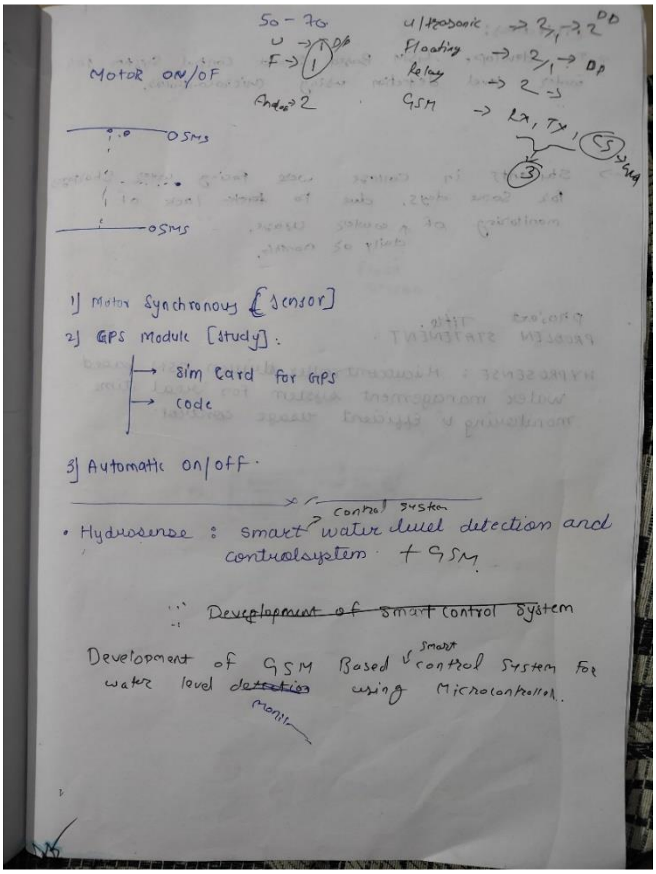
Sr. No.	Member Name	Work allotted	Work completed
1	Ms. Ritika P. Bhandari	Research Work + Code + Documentation	
2	Mr. Yash C. Sonje	Research Work + Code + Documentation	
3	Mr. Mayur M. Vaidya	Code + Hardware + Research Work	

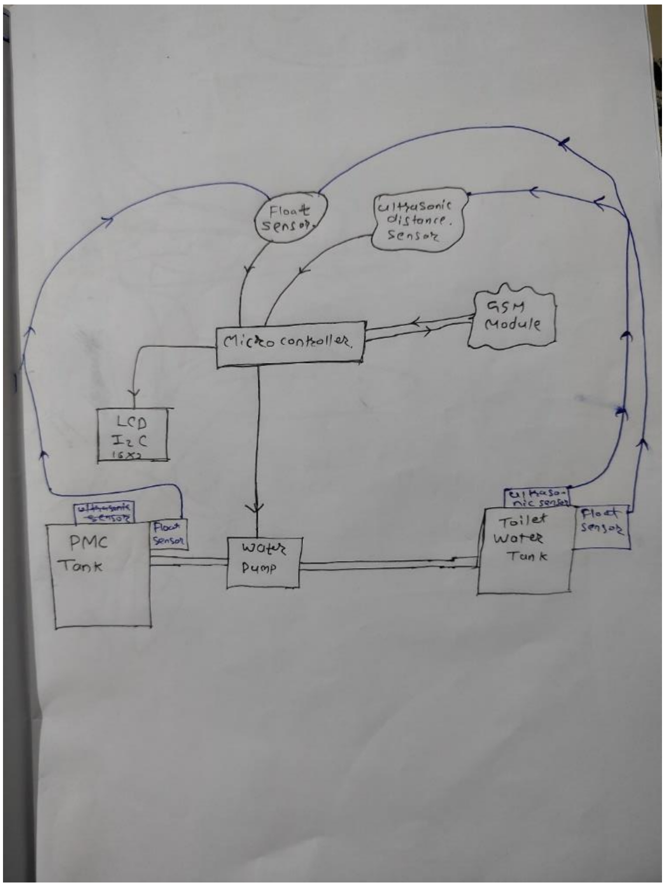
Chapter : 6 Datasheets

- https://docs.arduino.cc/resources/datasheets/A000005-datasheet.pdf
- https://filprosensors.com/assets/img1/filpro_pdf/Magnetic-Float-switch.pdf

Chapter : 7 Design Thinking








```
{
Pin mode (4, Input);
Pin mode (
~~Pin mode (LED_BUILTIN, OUTPUT);~~
} Pin mode (                    ) ;
// loop fun.
Void loop()
{
if (digital Read (4) ==0) & (digit R
```

CODE.

1. ultrasonic
2. LCD
3. GSM *
4. Relay
5. magnetic floating sensor + Rela
6. Tank (water data Usage) *
7. SD card *

- mag. floating sensor.
- SD card.

Objectives:

→ To develope GSM Based Smart Control System for water level detection using microcontroller.

→ Students in college were facing water shortages for some days, due to ~~fack~~ lack of monitoring of water usage.
daily or monthly,

Project Title.
PROBLEM STATEMENT.

HYDROSENSE : Microcontroller driven GSM based water management system for real time monitoring & efficient usage control.